

Product Dissection & Relational Data Architecture for Paisabazaar

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1. Company Overview

Paisabazaar, founded in 2014, is India's largest online platform for loans and credit cards, acting as a bridge between customers looking for credit and more than 50 partner banks and NBFCs. The platform solves the core challenge of fragmented product comparison by allowing users to compare personal loans, credit cards, home loans, and business loans on a single screen instead of visiting multiple bank branches or websites.

To help users stay informed about their financial health, Paisabazaar provides free monthly credit score tracking from all four RBI-regulated credit bureaus: CIBIL, Experian, CRIF High Mark, and Equifax. It also features a smart recommendation system called **"Chance of Approval,"** which uses an algorithm to match users with products they qualify for before they submit a formal application, preventing unnecessary rejections and protecting their credit scores. Additionally, the platform offers utility bill payments, recharges, and credit advisory tools to keep customers engaged and build a continuous picture of user financial behavior.

2. Product Dissection and Real-World Problems Solved

Paisabazaar simplifies borrowing and managing money by solving four major everyday problems:

1. **Comparing Loans & Credit Cards in One Place**
    - **Problem:** Checking loan rates across multiple bank websites and branches takes too much time and effort.
    - **Solution:** Paisabazaar brings 50+ banks and NBFCs together on one screen so users can quickly compare and pick the best option.
2. **Knowing Approval Odds Before Applying**
    - **Problem:** Applying for a loan without knowing if you qualify can lead to rejection, which damages your credit score.
    - **Solution:** The **"Chance of Approval"** tool uses AI to show your likelihood of getting approved before you apply, along with pre-approved offers to avoid unnecessary rejections.
3. **Free & Easy Credit Score Tracking**
    - **Problem:** Most people don't know their credit score or how missed payments affect their ability to get loans.
    - **Solution:** Users get free monthly credit reports from all 4 bureaus (CIBIL, Experian, CRIF, Equifax) and a **Score Simulator** to see how to improve their score.
4. **All-in-One Utility Payments**
    - **Problem:** Managing bills, recharges, and credit card payments across different apps is inconvenient.
    - **Solution:** An integrated bill payment section lets users pay utility bills and recharge in one app while earning cashback and staying engaged.

3. Case Study: Real-World Problems and Paisabazaar's Solutions

Problem 1: Too Many Bank Websites to Compare Loans

- **The Problem:** Visiting 10 different bank websites or branches to find interest rates takes too much time and creates confusion.
- **Paisabazaar's Solution:** It gathers offers from **50+ banks** on one screen so you can compare interest rates and fees side-by-side instantly.

Problem 2: Fear of Rejection and Credit Score Damage

- **The Problem:** Applying for a loan without knowing if you qualify can lead to rejection, which harms your credit score.
- **Paisabazaar's Solution:** The **"Chance of Approval"** tool uses AI to show your exact odds of getting approved *before* you apply, preventing unnecessary rejections.

Problem 3: Not Knowing How Credit Scores Work

- **The Problem:** Most people don't know their credit score or how late payments affect their ability to get future loans.
- **Paisabazaar's Solution:** Provides **free monthly credit reports** from all 4 bureaus (CIBIL, Experian, CRIF, Equifax) and a **Score Simulator** to help you improve your score.

Problem 4: Managing Bills Across Multiple Apps

- **The Problem:** Using separate apps for mobile recharges, electricity bills, and credit card payments is inconvenient.
- **Paisabazaar's Solution:** An **all-in-one bill payment section** lets you pay utility bills directly in the app while earning cashback.

4. Top Features of Paisabazaar

1. Free Multi-Bureau Credit Score Check

- **What it does:** Provides users with free monthly credit reports from all 4 RBI-regulated credit bureaus in India—**CIBIL, Experian, CRIF High Mark, and Equifax**.
- **Key Tools:** Includes a **Score Simulator** that lets users see how financial actions (e.g., paying off debt, closing a credit card) will impact their credit score before taking action.

2. Multi-Lender Product Marketplace

- **What it does:** Serves as a single digital marketplace aggregating loans and credit cards from over **50+ partner banks, NBFCs, and fintech lenders**.
- **Products Offered:** Personal loans, credit cards, home loans, business loans, car loans, and education loans with side-by-side comparison of interest rates and processing fees.

3. AI-Powered "Chance of Approval"

- **What it does:** Uses proprietary machine learning algorithms to match user financial profiles against partner lender eligibility rules.
- **Impact:** Shows users their exact likelihood of approval (*High, Medium, Low*) **before** they submit a formal application, preventing hard inquiries and score damage from rejections.

4. Pre-Qualified & Pre-Approved Offers

- **What it does:** Displays curated, soft-approved credit offers tailored specifically to the user's credit profile and income.
- **Impact:** Enables faster disbursals with minimal documentation and removes the risk of blind loan applications.

5. Personalised Credit Advisory Services

- **What it does:** Offers tailored recommendations and expert guidance for users looking to build, repair, or improve their credit profile.
- **Target Audience:** Especially valuable for "New-to-Credit" (NTC) individuals and borrowers recovering from past defaults.

6. Integrated Bill Payment & Utility Hub

- **What it does:** A unified payment module supporting mobile recharges, utility bill payments (electricity, gas, water), and credit card payments across 30+ categories.
- **Business Purpose:** Drives frequent user engagement while earning cashback rewards, creating recurring behavioral data to improve credit advisory models.

5. Schema Design Based on Top Features

To support Paisabazaar's core product capabilities—multi-bureau credit score tracking, multi-lender comparison, AI-driven approval estimates, application tracking, bill payments, and credit advisory—the relational database is structured into 8 core entities:

1. User Entity

Field	Type	Description
`UserID` **(PK)**	INT	Unique identifier for each registered user.
Full_Name	VARCHAR	User's full legal name.
Email	VARCHAR	Email address for communications and offer alerts.

Phone	VARCHAR	Mobile number used for OTP authentication.
City	VARCHAR	Residential city for location-based offer filtering.
Registration_Date	DATE	Account creation timestamp.

2. CreditScoreReport Entity

Field	Type	Description
`ReportID` **(PK)**	INT	Unique identifier for each credit pull.
`UserID` **(FK)**	INT	References the `User` table.
Bureau_Name	VARCHAR	Credit bureau (CIBIL, Experian, CRIF, Equifax).
Score	INT	Credit score value returned by the bureau.
Report_Date	DATE	Timestamp of when the report was fetched.

3. Lender Entity

Field	Type	Description
`LenderID` **(PK)**	INT	Unique identifier for each partner bank/NBFC.
Lender_Name	VARCHAR	Name of the bank or financial institution.
Lender_Type	VARCHAR	Entity classification (Bank, NBFC, Fintech).
RBI_Reg_ID	VARCHAR	Regulatory license registration reference.

4. Product Entity

Field	Type	Description
`ProductID` **(PK)**	INT	Unique identifier for each credit product.
`LenderID` **(FK)**	INT	References the `Lender` table.
Product_Type	VARCHAR	Personal Loan, Credit Card, Home

		Loan, etc.
Interest_Rate	DECIMAL	Annual percentage rate (APR).
Processing_Fee	DECIMAL	Upfront fee charged upon approval/disbursal.

5. Offer Entity

Field	Type	Description
`OfferID` **(PK)**	INT	Unique identifier for each generated offer match.
`UserID` **(FK)**	INT	References the matched `User`.
`ProductID` **(FK)**	INT	References the matched `Product`.
Chance_Of_Approval_Score	DECIMAL	AI-computed approval likelihood percentage.
Pre_Qualified_Flag	BOOLEAN	Flags whether the user is pre-qualified/pre-approved.
Offer_Date	DATE	Timestamp of when the offer was generated.

6. Application Entity

Field	Type	Description
`ApplicationID` **(PK)**	INT	Unique identifier for each submitted application.
`UserID` **(FK)**	INT	References the applying `User`.
`OfferID` **(FK)**	INT	References the specific `Offer` selected.
Application_Status	VARCHAR	Status (Submitted, Under Review, Approved, Disbursed).
Application_Date	DATE	Timestamp of formal application submission.
Disbursed_Amount	DECIMAL	Final sanctioned or disbursed loan amount.

7. BillPayment Entity

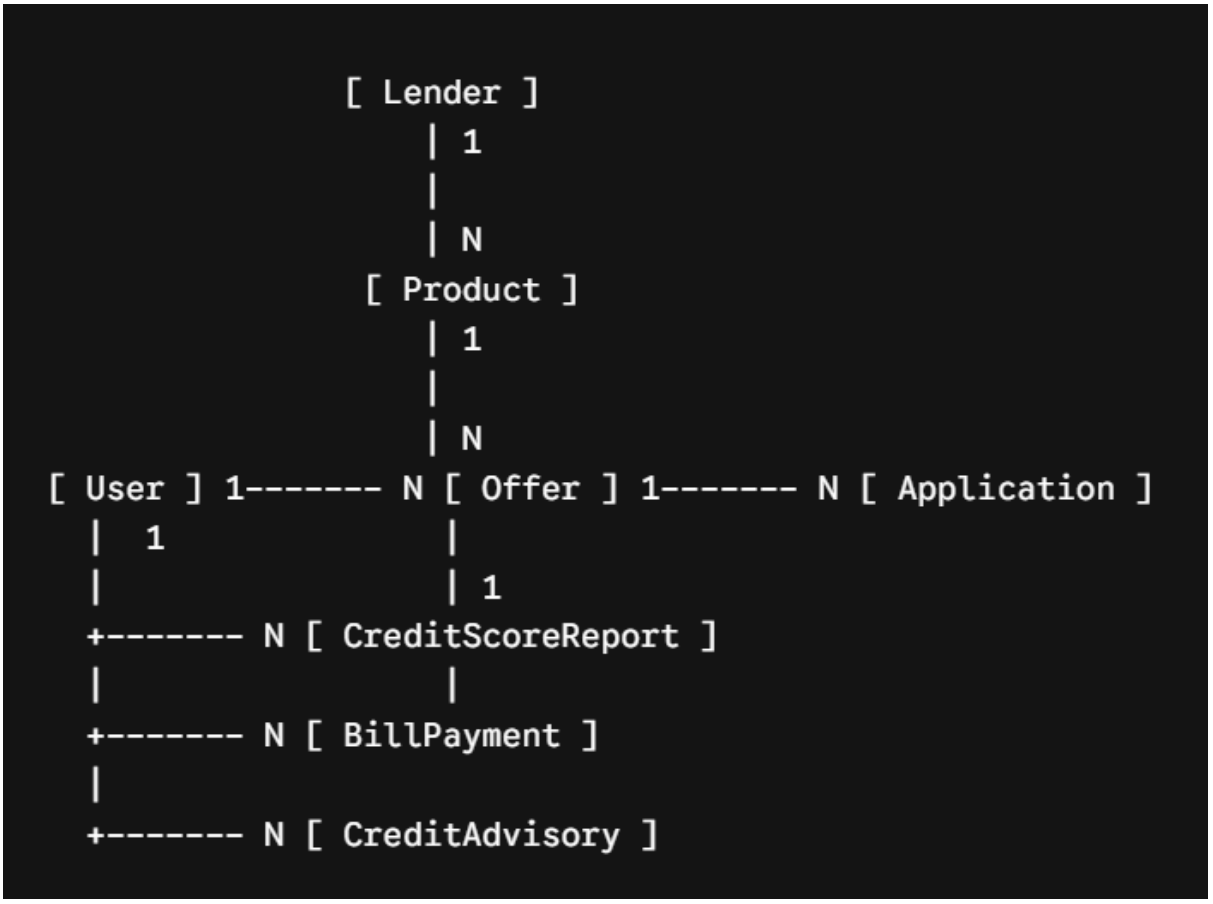
Field	Type	Description
`PaymentID` **(PK)**	INT	Unique transaction identifier.
`UserID` **(FK)**	INT	References the `User` making the

		payment.
Category	VARCHAR	Mobile, Electricity, Broadband, Credit Card Bill, etc.
Amount	DECIMAL	Total payment amount processed.
Cashback_Earned	DECIMAL	Rewards credited to user wallet for engagement.

8. CreditAdvisory Entity

Field	Type	Description
`AdvisoryID` **(PK)**	INT	Unique identifier for each advisory log.
`UserID` **(FK)**	INT	References the receiving `User`.
Advisory_Type	VARCHAR	Guidance type (Score Improvement Plan, Debt Reduction).
Recommendation_Date	DATE	Timestamp of generated recommendation.

6. Relationships



1. User ↔ CreditScoreReport (One-to-Many / 1:N)

- **Relationship:** One `User` can have multiple `CreditScoreReport` records over time, but each credit report belongs to exactly one `User`.
- **Business Logic:** Since credit reports are updated monthly across 4 bureaus (CIBIL, Experian, CRIF, Equifax), a single user accumulates historical report entries over time.
- **Foreign Key:** `CreditScoreReport.UserID` → `User.UserID`

2. Lender ↔ Product (One-to-Many / 1:N)

- **Relationship:** One `Lender` (Bank/NBFC) can offer multiple `Product` items, but each product is offered by exactly one `Lender`.

- **Business Logic:** HDFC or ICICI Bank (Lender) offers multiple financial products such as Personal Loans, Home Loans, and Credit Cards on Paisabazaar.
- **Foreign Key:** `Product.LenderID` → `Lender.LenderID`

3. User ↔ Offer ↔ Product (Many-to-Many via Offer / N:M)

- **Relationship:** A `User` receives multiple personalized `Offer` options, and a single `Product` can be offered to multiple users. The `Offer` table acts as a bridge entity between `User` and `Product`.
- **Business Logic:** Paisabazaar's AI recommendation engine generates dynamic offers for users based on their credit score and income, appending a calculated `Chance_Of_Approval_Score`.
- **Foreign Keys:**
  - `Offer.UserID` → `User.UserID`
  - `Offer.ProductID` → `Product.ProductID`

4. Offer ↔ Application (One-to-Many / 1:N)

- **Relationship:** One `Offer` can lead to multiple `Application` attempts (e.g., re-applying if documentation fails), but each application originates from a specific matched `Offer`.
- **Business Logic:** When a user clicks "Apply Now" on a pre-qualified offer, a formal application record is instantiated to track approval and disbursal stages.
- **Foreign Keys:**
  - `Application.OfferID` → `Offer.OfferID`
  - `Application.UserID` → `User.UserID`

5. User ↔ BillPayment (One-to-Many / 1:N)

- **Relationship:** One `User` can perform multiple `BillPayment` transactions, but each payment record belongs to only one `User`.
- **Business Logic:** Users routinely pay mobile, utility, and credit card bills within the platform to earn cashback, creating recurring behavioral transactional logs.
- **Foreign Key:** `BillPayment.UserID` → `User.UserID`

6. User ↔ CreditAdvisory (One-to-Many / 1:N)

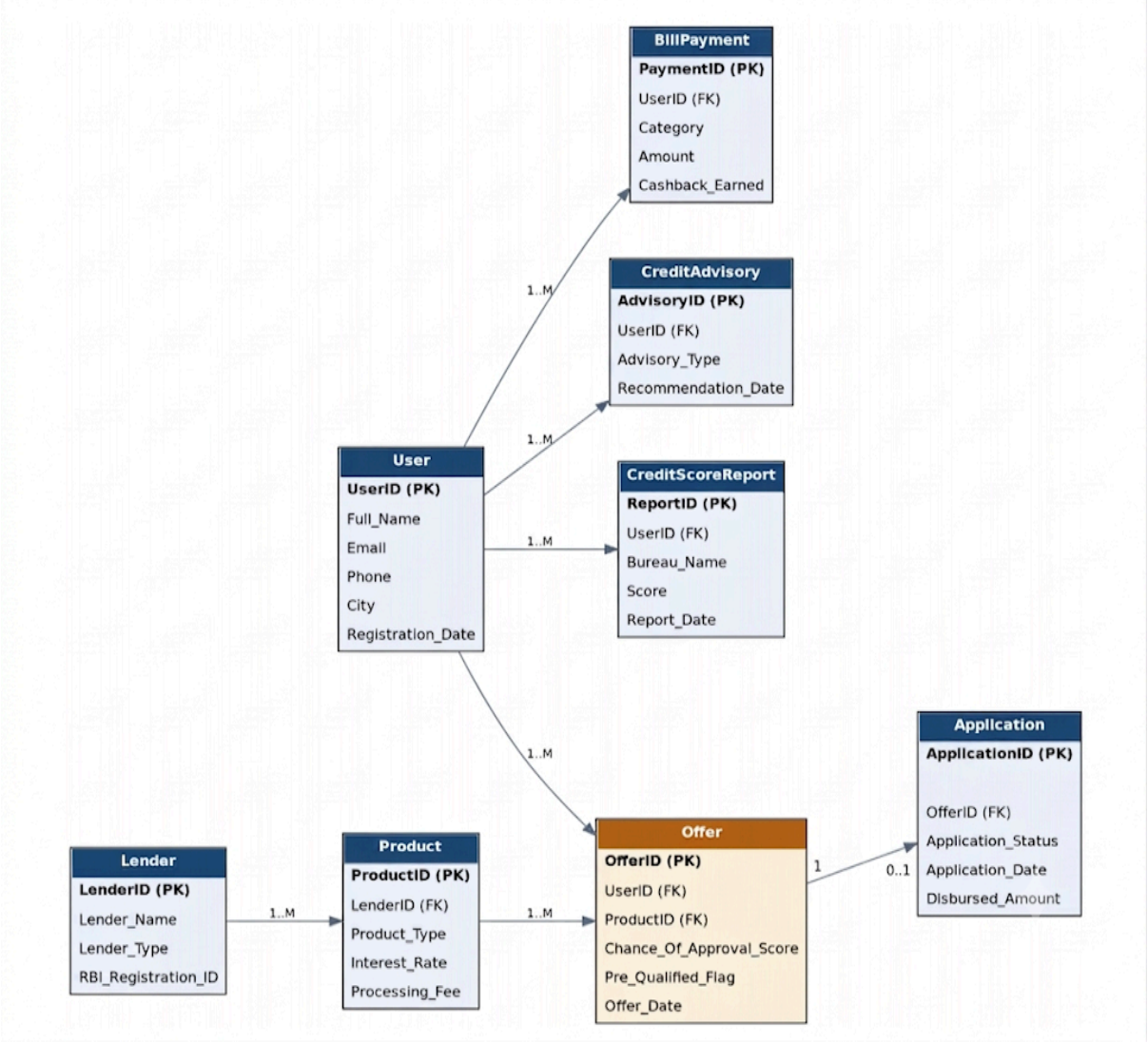
- **Relationship:** One `User` can receive multiple `CreditAdvisory` recommendations, but each advisory entry is linked to one specific `User`.
- **Business Logic:** Tailored score-improvement plans or debt-reduction strategies are generated periodically as the user's credit profile changes.
- **Foreign Key:** `CreditAdvisory.UserID` → `User.UserID`

7. Rationale Behind the Design

The schema is deliberately built around the User as the central entity, since every Paisabazaar feature — score tracking, offer matching, applications, bill pay, advisory — is personalised to one profile. CreditScoreReport is kept separate from User (rather than as columns on the User table) because a user has multiple bureau scores over time, and the platform's core promise is tracking score movement monthly, not just storing a single current value. Offer is modelled as its own entity rather than a direct User-Product link because it carries decision-relevant attributes (Chance_Of_Approval_Score, Pre_Qualified_Flag) that only exist at the moment of matching — this is what lets the platform show approval likelihood before an application is filed, which is the feature that most directly reduces failed applications and score damage. Application is kept separate from Offer so that the same offer can be tracked through itsfull lifecycle (submitted, under review, approved, disbursed) without

overwriting the original offer terms. Lender and Product are split apart because a single bank offers many products with different terms, and the marketplace's comparison feature depends on being able to query products independently of which lender they belong to. Finally, BillPayment and CreditAdvisory are kept as lightweight, append-only logs tied to User, reflecting their role as supporting engagement features rather than core lending workflows.

8. Entity-Relationship Diagram



This ER diagram illustrates the complete data model of a financial aggregation platform like Paisabazaar, depicting how user activity, credit evaluation, bank products, and loan workflows connect within

a unified system. At the center is the `User` table, which acts as the core anchor linking a single user to their multi-bureau credit score reports, utility bill payments, and credit advisory recommendations in a one-to-many relationship. On the institutional side, the `Lender` table connects to various financial `Product` offerings such as loans and credit cards. Rather than allowing direct applications that could harm a user's credit score through hard inquiries, the system uses an `Offer` bridge entity to algorithmically match users with suitable products and compute pre-approval scores. Finally, when a user chooses to proceed with an offer, an `Application` record is created to track the lifecycle of the loan from submission through final disbursal, ensuring complete normalization and operational traceability across the platform.

9. Conclusion

This database architecture provides a robust, scalable foundation for Paisabazaar’s credit marketplace by harmonizing smart technical engineering with business objectives. By algorithmically evaluating user eligibility through pre-screened offers prior to formal submission, the system protects customer credit scores from hard inquiries and high rejection rates. The normalized relational design eliminates data redundancies—such as transitive dependencies in application tracking—while uniting multi-bureau credit reports, recurring bill payments, and personalized advisory tools into a cohesive 360-degree financial profile. Furthermore, strict relational links between lenders, products, offers, and final disbursals give complete operational transparency for auditing bank commission payouts. Ultimately, this balance of user credit protection, normalized data design, and monetizable tracking delivers a performant platform built for seamless scaling and long-term financial aggregator success.